Statistics II: Introduction to Inference

Problem set 2

1. Let $Y \sim \text{Exponential}(\lambda)$. Find the distribution of $X = Y^k$ for $k > 0$. Find the expectation of X.

2. Let $F_{\boldsymbol{\theta}}$ be $\texttt{Binomial}(m, \rho)$ where $\theta = \rho$.

   (a) Consider the class of linear estimators of ρ, i.e., the estimators of the form $T_{\mathbf{l}}(\mathbf{X}) = \sum_{j=1}^{n} l_j X_j$, $l_j \in \mathbb{R}$, $j = 1, \ldots, n$. Find conditions on $\mathbf{l} = (l_1, \ldots, l_n)'$ under which $T_{\mathbf{l}}(\mathbf{X})$ is an unbiased estimator of ρ.

   [*Let* $\mathbf{l}^\star$ *be a choice of* $\mathbf{l}$ *which satisfies the condition obtained in part (a). Then the estimator obtained by replacing* $\mathbf{l}$ *by* $\mathbf{l}^\star$ *in* $T_{\mathbf{l}}(\mathbf{X})$ *is called a* linear unbiased estimator *of* ρ.]

   (b) Find the variance of $T_{\mathbf{l}}(\mathbf{X})$. Denote it by $\sigma_{\mathbf{l}}^2$.

   (c) Minimize $\sigma_{\mathbf{l}}^2$ with respect to $\mathbf{l}$ subject to the constraint obtained in part (a).

   [*Let the solution obtained in part (c) be* $\mathbf{l}^\star$. *The estimator obtained by replacing* $\mathbf{l}$ *by* $\mathbf{l}^\star$ *in* $T_{\mathbf{l}}(\mathbf{X})$ *is called the* Best Liner Unbiased Estimator (BLUE).]

   (d) Is the BLUE same as the UMVUE of ρ?

3. Let $F_{\boldsymbol{\theta}}$ be $\texttt{normal}(\mu, \sigma^2)$ and $\boldsymbol{\theta} = (\mu, \sigma^2)$. Consider the class of estimators of the form

$$S_{a_n}^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}{a_n}, \quad a_n > 0.$$

   Find the estimator which minimizes the MSE in estimating σ^2 in this class. What is the bias associated with the best estimator?

4. Let $X_1, \ldots, X_n \stackrel{i.i.d.}{\sim} \texttt{Uniform}(0, \theta)$.

   (a) Is $X_{(n)} = \max\{X_1, \ldots, X_n\}$ an unbiased estimator of θ? Can you find a function of $X_{(n)}$, say $T_n = h(X_{(n)})$, which is an unbiased estimator of θ.

   (b) Find the distribution of $Z_{n,\theta} = (X_{(n)} - \theta)^2$. Hence, or otherwise find the MSE of $X_{(n-1)}$ in estimating θ.

   (c) Find the MSE of T_n in estimating θ.

   (d) Consider another estimator of θ, $S_n = 2\bar{X}_n$. Find the S_n in estimating θ.

   Among these three estimators $X_{(n)}$, T_n and S_n, which estimator has the lowest MSE?

5. The waiting time (in whole minutes) for a bus is distributed as a $\text{Poisson}(\lambda)$ distribution. We are interested in estimating the probability that the waiting time is at least one minute, i.e.,

$$\psi(\lambda) = P(X \geq 1) = 1 - \exp(-\lambda).$$

   To estimate this probability, we propose collecting the waiting times of n individuals, denoted as $X_1, \ldots, X_n$. Assuming $X_1, \ldots, X_n$ are independently distributed, find an unbiased estimator of $\psi(\lambda)$.